SĀVI: Wireless BCI-Based Emergency Detection & Response

"Seconds Matter"

CONFIDENTIAL PROJECT PROPOSAL Advanced Neuro-Safety & Automated Intervention Platform

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1. Executive Summary

SĀVI represents a paradigm shift in emergency medical monitoring through the fusion of neuroscience and IoT driven response systems. Traditional emergency wearables suffer from a critical limitation: they rely on manual triggers (panic buttons) or reactive motion sensors, often failing when a patient is incapacitated during a seizure or unconscious following a fall.

Our proprietary **Neural-Motion Logic** addresses this gap by synchronizing a Wireless Brain-Computer Interface (BCI) with high-precision motion analysis. By monitoring 6-channel EEG data and 6-axis IMU movement, SĀVI detects emergencies at the neurological source, enabling a response time up to **85% faster** than manual systems. With a multi-tier escalation matrix and GSM-fallback for offline reliability, SĀVI ensures help reaches the patient even if family members are unavailable or network connectivity fails.

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2. Problem Statement: The Emergency Response Vacuum

Current monitoring protocols for epilepsy and elderly care are fundamentally incomplete, leading to preventable complications:

- **The Incapacitation Gap:** Seizures and high-impact falls often leave patients unable to call for help. Existing systems that lack automated BCI triggers create a false sense of security.
- **Connectivity Dependency:** Standard "smart" alarms fail in low-network areas. A lost Wi-Fi signal currently represents a total failure in patient safety.
- **The Verification Blind Spot:** Caregivers often receive alerts without visual context, leading to delayed prioritization or high false-alarm fatigue.
- **Structured Escalation Failure:** There is currently no robust mechanism to bridge the gap if the primary caregiver is unreachable, leaving the patient in a "response vacuum" during the critical Golden Hour.

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3. The SĀVI Solution: System Architecture

3.1 Hardware Fusion Layer

SĀVI utilizes a clinical-grade biosensor suite integrated with the **ESP32-S3** platform to capture multi-modal data streams:

Category	Component	Specification / Requirement	Purpose & Strategic Function
Core Processing	ESP32-S3 DevKit	Dual-core, 16MB Flash, AI/TinyML acceleration	Central hub for Edge-AI seizure classification and multi-modal sensor fusion.
Neuro-Sensing	ADS1292R / ADS1299	24-bit Analog Front-End (AFE)	Medical grade EEG acquisition to detect abnormal neural spikes.
Motion Detection	MPU6050 / BMI270	6-Axis IMU (Accel + Gyro)	Real-time fall detection and identification of tonic-clonic motor patterns.
Visual Context	ESP32-CAM	OV2640 2MP Camera Sensor	Triggered visual verification to reduce caregiver panic and false alarms.
Connectivity	SIM800L GSM	2G/GPRS Quad-band with SMS Fallback	Ensures emergency alert delivery in internet-dead zones or rural areas.
Patient Interface	Snap-on Electrodes	Disposable Ag/AgCl or Dry Active Pins	Secure physiological signal capture with low impedance.
Power System	TP4056 + Li-Po	3.7V 1000mAh+ with Protection Circuit	Reliable, rechargeable energy source capable of 2A GSM bursts.
Feedback System	LRA / Buzzer	Haptic Motor & 5V Active Buzzer	Discreet "Pre-Alert" vibration and loud "Tier 3" audible emergency alarms.
App Framework	Flutter & Firebase	Real-time NoSQL Database	Cross-platform dashboard for live tracking and caregiver escalation.
AI Architecture	TinyML / Python	Convolutional Neural Network (CNN)	On-device classification of "Normal" vs. "Seizure" states.

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3.2 The SĀVI Logic: Core Innovation

Our proprietary algorithm performs continuous state classification to differentiate between daily activity and life-threatening events:

Emergency Index = f(EEG Spikes, IMU Impact, Duration)

Escalation States:

- **LOW (Normal):** Continuous background monitoring; low-power state.
- **MODERATE (Potential Event):** Haptic vibration alert to user; local data logging begins.
- **HIGH (Confirmed Emergency):** Immediate trigger of the Tiered Response Matrix.

3.3 Tiered Response Matrix

1. **Tier 1 (Instant):** Push notification + Alert to primary family with GPS and live photo via Flutter App.
2. **Tier 2 (T + 5 Mins):** If unacknowledged, automated GSM-SMS sent to neighbours and secondary contacts.
3. **Tier 3 (T + 7 Mins):** Automatic escalation to local healthcare providers and emergency services.

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4. Software & Data Infrastructure

- **Edge-AI (TinyML):** Processing happens on the **ESP32-S3** to ensure zero-latency detection and patient data privacy.
- **Data Structure:** Utilizes high-efficiency **Circular Buffers** for EEG streaming and **Event-Driven Structs** for the escalation state machine.
- **Cross-Platform App:** A Flutter-based interface provides caregivers with real-time health dashboards and a one-tap "Emergency Verified" button.

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5. Market Analysis and Scalability

- **Market Opportunity:** The global BCI market is projected to reach **$3.3 billion by 2026**, with the non-invasive healthcare segment being the fastest-growing vertical.
- **Addressable Population:** 50 million+ epilepsy patients worldwide and an aging global population requiring at-home fall monitoring.

- **Phased Scaling:**
    - **Phase 1:** Pilot launch for epilepsy clinics and home-care providers.
    - **Phase 2:** Integration with national emergency databases and elderly care facilities.
    - **Phase 3:** Expansion into high-risk industrial monitoring (e.g., lone-worker safety).

6. Expected Impact

- **Reduced Response Time:** Targeted 60% reduction in time-to-treatment for seizure events.
- **Operational Reliability:** 99.9% alert delivery success rate through redundant GSM/Wi-Fi channels.
- **Peace of Mind:** Eliminates constant manual checking for caregivers through autonomous "Always-On" monitoring.

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7. Conclusion

SĀVI represents the convergence of neurotechnology, IoT, and compassionate design. By making the invisible visible revealing the real-time electrical state of the brain we empower families to protect their loved ones even when they are physically apart. The current gap in emergency care exists because we measure only half the equation. SĀVI completes the picture.

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